Experimental Validation of AUX scheme for Quantum Homomorphic Encryption on IBM Quantum Platforms

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***Abstract*—Quantum Homomorphic Encryption (QHE) addresses Quantum Cloud Computing (QCC) security concerns by ensuring the privacy of a client's data and algorithms when outsourced to untrusted third-party quantum servers. However, current QHE schemes face significant challenges: scaling computational resources introduces overhead and hardware noise, degrading accuracy and compromising security. This paper implements and analyses a non-interactive AUX-QHE scheme that employs pre-generated auxiliary states for universal computation. We identify three critical computational bottlenecks: exponential growth in auxiliary state count, complex homomorphic evaluation, and extensive symbolic key updates. Through experimental evaluation on IBM Quantum hardware, we quantify the impact of NISQ noise on AUX-QHE performance and establish practical resource thresholds for deployment. Our results bridge the gap between theoretical QHE frameworks and their practical implementation on noisy quantum devices, providing concrete benchmarks for future noise mitigation efforts.**

***Index Terms*—Quantum Cloud Computing (QCC), Quantum Homomorphic Encryption (QHE), Auxiliary (AUX), Quantum Error Correction (QEC), Noisy Intermediate-scale Quantum (NISQ)**

I. Introduction

Existing QCC is a hybrid model that connects classical cloud infrastructure to current quantum hardware, enabling clients to remotely access quantum computing without local hardware [1]. However, this QCC architecture could expose sensitive data during delegated computation, posing serious challenges for high-data confidentiality fields such as defence, healthcare, and finance [2]. The QHE concept is recognised as a secure computation protocol, as it allows a quantum server to compute on encrypted data without decrypting the initial quantum states [3], [4].

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Homomorphic evaluation over encrypted quantum states efficiently handles Clifford gates. However, evaluating non-Clifford gates, which are essential for achieving Quantum Fully Homomorphic Encryption (QFHE) and universal quantum computation, introduces a theoretical fundamental challenge: unexpected Phase error [4]. Yu et al. show that the no-go theorem prevents meeting all of QFHE's conditions: universality, compact decryption, information-theoretic security, and non-interactive operation. As a result, each practical QHE algorithm must sacrifice one of these conditions. This leads to high computational overhead and substantial hardware requirements, which limit practical deployment [3].

Broadbent and Jeffery (2015) [5] introduced a non-interactive QHE scheme. They integrated this with a classical Fully Homomorphic Encryption (FHE) scheme to address no-go theorem limitations. The FHE-QHE scheme has two evaluation methods: AUX and Einstein-Podolsky-Rosen (EPR). Both aim to fix phase errors from T-gate operations during quantum computation [5]. The EPR uses Bell pair entanglement to delay T-gate errors. The AUX prepares correction tools in advance.

Building on these foundations, Fernandez and Martin-Delgado [6] simulated the EPR approach for Grover's algorithm in the Qiskit environment, revealing that the algorithm succeeds only with three qubits and at most 7 (T/$T^\dagger$) gates. Similarly, Ganjian et al [7] demonstrated the FHE-QHE scheme using the EPR evaluation method, demonstrating a practical threshold of 6 T-gates. Across these implementations, the theoretical and experimental EPR schemes introduce algorithmic computation noise and become impractical with many T-gates.

For AUX, generating auxiliary qubits quickly becomes theoretically infeasible at high T-depth [5]. Alagic et al. [8] proposed a vQFHE scheme, TrapTP, that leverages a classical FHE scheme to mitigate the additional overhead associated with auxiliary inputs. However, this vQFHE scheme exhibits an exponential increase in the complexity of evaluation keys, which

requires large-scale quantum infrastructure. Mahadev [9] proposed an alternative approach that uses a classical FHE method to handle T-gate computations, replacing the quantum EPR or AUX. Although this approach can evaluate circuits with a polynomial number of T-gates, it introduces large ciphertexts and significant overhead in classical computation. Despite these theoretical advances and significant exploration of AUX-QHE and related QHE variants, no practical FHE-AUX-QHE framework has yet been implemented on actual Noisy Intermediate-scale Quantum (NISQ) systems. Existing studies offer limited practical noise-adaptation strategies or detailed resource analyses for the AUX method. This research fills these gaps by implementing and evaluating FHE-AUX-QHE on quantum hardware, assessing its practicality in real-world noise environments, and establishing benchmarks for noise mitigation and resource requirements. Our contributions are as follows:

- We design and implement all complex features of the AUX-QHE algorithm, including pre-generated auxiliary qubits and symbolic key management. These elements establish a practical foundation for AUX methods, enabling precise identification of computational overheads.
- By analysing the performance of the scheme on the IBM Quantum platform, we assess the impact of noisy hardware, which helps understand the feasibility of AUX and EPR in practice.

These strategies are vital for understanding scalability factors toward practical QHE and ensuring its compatibility with future QFHE models.

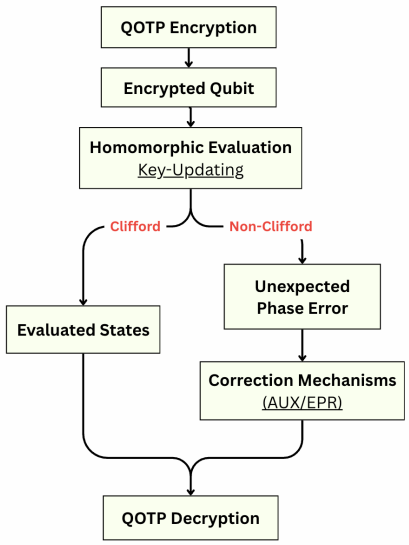
The remainder of this paper is organised as follows. Section 2 reviews related work and gaps in the AUX method in QHE algorithms. Section 3 outlines the theoretical background and evaluation methodology. Section 4 describes the algorithm's implementation on IBM Quantum. Section 5 describes the experimental approach. Sections 6 and 7 analyse results and discuss implications for scalability and feasibility.

II. Theoretical Background

A. Architecture of QHE Algorithms

The QHE scheme, particularly when combined with classical FHE as described by Broadbent and Jeffery [5], is typically composed of four core algorithms: Key Generation (QHE.KeyGen), Encryption (QHE.Enc), Homomorphic Evaluation (QHE.Eval), and Decryption (QHE.Dec). Initially, the client uses Quantum One-Time Pad (QOTP) to encrypt quantum information by applying random Pauli-X and Pauli-Z operations, determined by secret keys [10], [11]. The server then performs quantum homomorphic evaluation (QHE.Eval), which must maintain encryption while correctly updating the QOTP keys to ensure proper decryption—a procedure known as key updating [12]. Nearly all QHE algorithms decompose into two parts: Clifford gate evaluation (X, Z, H, P, and CNOT gates) and non-Clifford gate evaluation (T-gates) [13]. The non-Clifford evaluation introduces an unexpected T-gate challenge, which becomes the primary obstacle in this QHE concept 1. Finally, the client recovers the updated keys and decrypts the QOTP encryption via inverse Pauli operations.

Fig. 1: Relationship between QOTP and Homomorphic Evaluation within QHE



B. Fundamental EPR And AUX Evaluation

Among the various non-interactive QHE schemes, the EPR and AUX methods are recognised as foundational approaches for universal quantum computations [3], [4], [9].

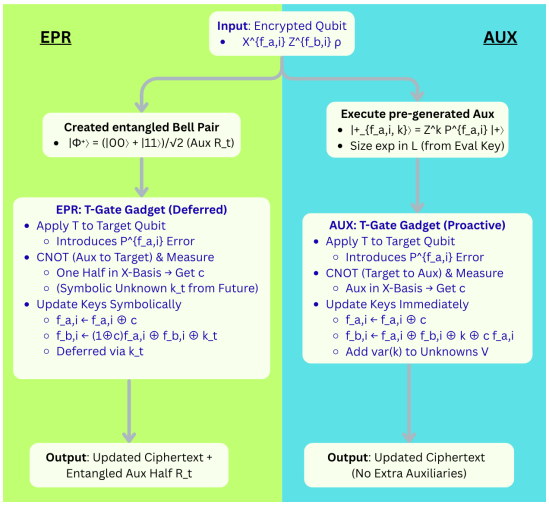
The EPR approach leverages quantum entanglement via Bell pairs to defer P-gate corrections until decryption, thereby shifting the computational burden to the decryption phase [5]. This method allows unrestricted T-gate arrangement and arbitrary quantum circuits in QHE algorithms. However, this deferral strategy leads to quadratic growth in decryption complexity, which severely limits the scalability of the EPR framework.

The AUX-based approach (AUX-QHE) offers an alternative by using pre-generated auxiliary qubits to handle T-gate evaluation, expressed as: $|+_{f,k}\rangle = Z^k P^f |+\rangle$ [5]. The AUX framework maintains compact decryption while proactively managing T-gate phase errors through specialised quantum circuits called T-gadgets, which are designed to control conditional P-gate errors. As shown in Fig 2, although the EPR and AUX evaluation methods both utilise T-gate gadgets, these gadgets are implemented differently due to distinct mechanisms for handling T-gates and key updates. In the AUX approach, T-gadgets consume auxiliary qubits to process T-gates homomorphically, effectively isolating phase errors and maintaining encryption security.

While the AUX approach offers more efficient decryption compared to EPR, it faces a critical scalability challenge: the number of required auxiliary states grows

exponentially with the circuit's T-depth. For a circuit with T-depth L and n as the number of qubits, the auxiliary resource requirement scales as $O(n^{2^{L-1}+1})$, making large-scale implementations computationally prohibitive [5].

Fig. 2: Differences between the EPR and AUX schemes



C. Hardware Noise Analysis

NISQ hardware faces significant challenges, including limited qubit counts (100 to 1000 qubits), significant error rates (gate fidelities approximately 99%), restricted computational depth and short qubit coherence times [14]. These constraints limit both the complexity of executable quantum algorithms and the implementation of Quantum Error Correction (QEC), which requires thousands of physical qubits per logical qubit [15]. Consequently, NISQ devices impose stringent resource and noise-tolerance requirements on quantum algorithms, particularly for resource-intensive schemes such as QHE.

For QHE specifically, computational overhead from non-Clifford evaluation amplifies hardware noise beyond simulated environments [16]. Since QEC remains infeasible on current hardware [17], Quantum Error Mitigation (QEM) focuses on mitigating hardware noise through post-processing of measurement results without modifying the underlying quantum circuit [18]. These QEM methods leverage the inherent error characteristics of quantum hardware and utilise statistical techniques, such as Probabilistic Error Cancellation (PEC) and Zero-Noise Extrapolation (ZNE), to mitigate noise effects [19]. Although QEM has significantly reduced errors, its effectiveness is constrained by fundamental noise levels and sampling overhead.

III. Implementation Methodology

We demonstrate the FHE-AUX-QHE framework with auxiliary-state management and validate its performance under hardware noise on the IBM Quantum platform [20]. The implemented algorithm consists of

Algorithm 1 AUX-QHE: Complete Protocol with Five Phases

Require: Circuit $\mathcal{C}$ (n qubits, T-depth L), security λ
Ensure: Homomorphically evaluated circuit with error mitigation

1: **// Phase 1: Key Generation**
2: $(pk, sk, evk) \leftarrow \text{BFV.KeyGen}(1^\lambda)$; $a, b \leftarrow \{0,1\}^n$; $\mathcal{V} \leftarrow \{a_i : a[i], b_i : b[i]\}_{i\in[n]}$; $\mathcal{T}_1 \leftarrow \{a_i, b_i\}_{i\in[n]}$; $\mathcal{A} \leftarrow \{\}$
3: $\forall \ell \in [2, L]$: $\mathcal{T}_\ell \leftarrow \mathcal{T}_{\ell-1} \cup \{tt' : t, t' \in \mathcal{T}_{\ell-1}\} \cup \{k_{i,\ell-1} : i \in [n]\}$
4: $\forall(\ell, i, t)$: $k \leftarrow \text{H}(\ell, i, t) \bmod 2$; $s \leftarrow t(\mathcal{V}) \bmod 2$; $\mathcal{A}[(\ell, i, t)] \leftarrow (Z^k P^s|+\rangle, s, k)$

5: **// Phase 2: Encryption**
6: $\mathcal{C}_{enc} \leftarrow X^a Z^b \cdot \mathcal{C}$; $\tilde{a}_i \leftarrow \text{BFV.Enc}_{pk}(a[i])$, $\tilde{b}_i \leftarrow \text{BFV.Enc}_{pk}(b[i])$ $\forall i$

7: **// Phase 3: Homomorphic Evaluation**
8: $f_a^{(i)} \leftarrow$ "a_i", $f_b^{(i)} \leftarrow$ "b_i" $\forall i$; $\mathcal{C}_{eval} \leftarrow \emptyset$; $\tau \leftarrow 1$
9: **for** each gate g in $\mathcal{C}_{enc}$ **do** ▷ Update key formulas per gate
10: H_i: swap$(f_a^{(i)}, f_b^{(i)})$; $\text{CNOT}_{i,j}$: $f_a^{(j)} \oplus\leftarrow f_a^{(i)}$, $f_b^{(i)} \oplus\leftarrow f_b^{(j)}$; S_i: $f_b^{(i)} \oplus\leftarrow f_a^{(i)}$
11: T_q: $(|\psi\rangle, s, k) \leftarrow \mathcal{A}[(\tau, q, f_a^{(q)})]$; $c \leftarrow \text{H}(f_a^{(q)}, k) \bmod 2$; $c_v \leftarrow$ "c"$_q_\tau$; $k_v \leftarrow$ "k"$_q_\tau$; $\mathcal{V}[c_v] \leftarrow c$; $\mathcal{V}[k_v] \leftarrow k$; $f_a^{(q)} \oplus\leftarrow c_v$; $f_b^{(q)} \leftarrow f_a^{(q)} \oplus f_b^{(q)} \oplus k_v$; if c=1: $f_b^{(q)} \oplus\leftarrow f_a^{(q)}$; $\tau \leftarrow \tau + 1$
12: $\mathcal{C}_{eval}$.add(g)
13: **end for**
14: $\tilde{a}'_i \leftarrow \text{BFV.Eval}_{evk}(f_a^{(i)}, \mathcal{V})$; $\tilde{b}'_i \leftarrow \text{BFV.Eval}_{evk}(f_b^{(i)}, \mathcal{V})$ $\forall i$

15: **// Phase 4: Decryption**
16: $a'_i \leftarrow \text{BFV.Dec}_{sk}(\tilde{a}'_i) \bmod 2$; $b'_i \leftarrow \text{BFV.Dec}_{sk}(\tilde{b}'_i) \bmod 2$ $\forall i$; $\mathcal{C}_{dec} \leftarrow \mathcal{C}_{eval} \cdot Z^{b'} X^{a'}$

17: **// Phase 5: Error Mitigation (ZNE)**
18: If ZNE: $\forall \lambda \in [1, 1.5, 2, 2.5]$: $\mathcal{D}$.add$((\lambda, \text{Execute}(\text{Amplify}(\mathcal{C}_{dec}, \lambda))))$; **return** $(\mathcal{C}_{dec}, \text{FitModel}(\mathcal{D}).\text{extrap}(0))$
19: **return** $\mathcal{C}_{dec}$

five phases: key generation, encryption, homomorphic evaluation, decryption, and error mitigation.

A. Key Generation and QOTP Encryption

The key generation phase (lines 2-4) initialises Brakerski-Fan-Vercauteren (BFV) keys and random QOTP keys. The term set $\mathcal{T}_\ell$ captures all possible key-polynomial expressions, while the variable dictionary $\mathcal{V}$ tracks all key-related values (lines 2,3). Pre-generated auxiliary states $|+_{f,k}\rangle$ are deterministically computed using hash functions (line 4). The encryption phase (line 6) then applies QOTP via Pauli operations and encrypts the QOTP keys using BFV.

B. Homomorphic Evaluation and Auxiliary Management

The homomorphic evaluation phase (lines 8-10) maintains encryption while updating symbolic key representations $f_a^{(i)}, f_b^{(i)}$ to enable proper decryption. Clifford gates update keys according to deterministic rules that preserve the QOTP structure (line 10), ensuring consistent symbolic key management throughout the process. When processing non-Clifford gates, the algorithm employs T-gadgets with pre-generated auxiliary states to handle phase errors (lines 11). For each T-gate, the auxiliary state is retrieved, a deterministic measurement outcome c is computed, and keys are updated, with conditional correction when $c = 1$ (line 11). After completing all gate operations, the symbolic

key polynomials are homomorphically evaluated using BFV (line 14), producing encrypted updated keys that account for all gate transformations.

C. QOTP Decryption and Error mitigation techniques

The decryption phase (line 16) recovers the updated keys using BFV and then removes the QOTP encryption via inverse Pauli operations. Error mitigation follows using ZNE (lines 18-19), which executes circuits at noise-scaling factors $\mathcal{A}$, fits exponential models, and extrapolates to the zero-noise limit. Additionally, IBM Quantum's circuit optimisation levels (0 and 3) are also evaluated to assess the impact of transpilation on noise exposure.

IV. Experimental Results

A. Experimental Setup

The FHE-AUX-QHE algorithm is executed in both local simulation (Table I) and quantum hardware (Table II). Our demonstration builds the classical symbolic computations and a quantum circuit via IBM Qiskit on a MacBook M3 Max with 36GB of RAM, using the IBM Quantum backend (ibm_{torino}) with 133 qubits [21].

We design three test-case circuits with configurations denoted as nq-Lt, where n is the number of qubits and L is the T-depth: 5q-2t, 4q-3t, 5q-3t, with (e.g., 5q-3t denotes a 5-qubit circuit with 3 T-gates). based on theoretical auxiliary scaling [5]. Each circuit includes single-qubit Clifford gates (H, S), CNOT gates for entanglement, and three randomly placed T gates. Algorithmic correctness is validated by using the fidelity F-score, comparing decrypted circuit outputs with unencrypted circuits. [5]. Moreover, we analyse resource overhead computational metrics, including the number of auxiliary states, the accumulated polynomial cross-product, and layer size.

B. Fidelity and Correctness Analysis

Table I confirms algorithmic correctness with perfect fidelity $(F = 1.0)$ for all configurations in simulation. By comparison, Table II reveals severe hardware noise degradation across all configurations. For 5q-2t, standalone ZNE reduces fidelity below the baseline $(F = 0.3569)$ to $(F = 0.3296)$, while Opt-3 $(F = 0.3618)$ and the combined Opt-3+ZNE method $(F = 0.3782)$ yield improvements. In the 4q-3t case, all three mitigation methods produce lower fidelities than the baseline $F = 0.1222$, suggesting that at higher T-depth, mitigation overhead may introduce additional noise rather than reducing it. For 5q-3t, ZNE alone produces the lowest fidelity $(F = 0.1175)$, while both Opt-3 $(F = 0.1293)$ and Opt-3+ZNE $(F = 0.1305)$ improve over the baseline $(F = 0.1199)$.

C. Algorithmic computational noise

Section B of Table I shows that auxiliary states increase from 575 (5q-2t) to 10,776 (4q-3t) and 31,025 (5q-3t), illustrating exponential growth as T-depth rises. The efficiency metric captures the growing number of auxiliary states per qubit layer, from 287.5 to 10,341.7. These figures show that auxiliary state growth becomes the dominant computational cost in QHE. Section C of Table I presents the corresponding operational times. For 5q-2t, auxiliary preparation takes only $0.009s$, yielding a total runtime of $0.042s$. For 5q-3t, auxiliary preparation increases to $0.597s$, accounting for 90.0% of the total runtime of $0.664s$. Notably, decryption time remains stable across all configurations $(0.031\text{–}0.032s)$, confirming that the AUX approach maintains compact decryption independent of circuit complexity.

D. Symbolic Key Complexity

Section B of Table I shows that symbolic QOTP key cross-terms grow rapidly with T-depth, increasing from 45 (5q-2t) to 5,505 (5q-3t) — a 121-fold increase. Layer sizes track $\mathcal{T}_\ell$ operations at each T-depth, growing from $(10, 105)$ at 5q-2t to $(10, 105, 6090)$ at 5q-3t, demonstrating exponential scaling. While classical BFV evaluation remains fast at low T-depths (Section C), the rapid growth of cross-terms at higher T-depths poses a risk of exceeding quantum decoherence time limits.

V. Discussion and Future Work

Hardware experiments demonstrate that the FHE-AUX-QHE algorithm faces fundamental constraints on current NISQ systems that are not captured by theoretical analysis, which assumes ideal noise-free operation.

A. NISQ Feasibility and Security Implications

The gap between simulation $(F = 1.0)$ and hardware fidelities $(F = 0.12\text{–}0.38)$ reveals both practical and security challenges for deploying QHE on NISQ devices. Our experiments identify two primary scalability bottlenecks: exponential growth in auxiliary states with increasing T-depth and rapid increases in symbolic key complexity. At the 5q-3t configuration, auxiliary-state requirements reach 31,025, with 5,505 cross-terms, representing the practical upper bound for current systems. From a security standpoint, the FHE-AUX-QHE scheme is proven q-IND-CPA secure [5], where QOTP provides information-theoretic encryption and BFV ensures computational key protection based on the Learning With Error (LWE) assumption. However, since QHE security fundamentally depends on correct computation over encrypted data, low hardware fidelity poses a direct threat to practical security since noise-induced errors in QOTP operations and key updates degrade decryption accuracy, undermining the guarantee of reliable encrypted computation on noisy hardware.

TABLE I: AUX-QHE Simulation Performance: Fidelity, Key Complexity, and Timing

	A. Correctness	B. Key & Auxiliary Complexity				C. Timing (s)			
Config	F	Aux States	Layer Sizes	Cross Terms	Efficiency	Aux Prep	T-Gadget	Decrypt	Total
5q-2t	1.0000	575	$10, 105$	45	287.5	0.009	0.001	0.032	0.042
4q-3t	1.0000	10776	$8, 68, 2618$	2306	3592.0	0.183	0.013	0.031	0.228
5q-3t	1.0000	31025	$10, 105, 6090$	5505	10341.7	0.597	0.036	0.032	0.664

TABLE II: Hardware Fidelity (F-score) Across Configurations and Optimisation Methods on IBM Quantum

Method	5q-2t	4q-3t	5q-3t
Baseline	0.3569	0.1222	0.1199
ZNE	0.3296	0.1162	0.1175
Opt-3	0.3618	0.1165	0.1293
Opt-3+ZNE	0.3782	0.1145	0.1305

Opt-3: IBM Qiskit transpiler optimisation level 3.
ZNE: Zero-Noise Extrapolation.

B. Error Mitigation and Future Directions

The combined error mitigation method (ZNE + Opt-3) offers improved fidelity over standalone methods across most configurations, though at 4q-3t, all mitigation methods underperform the baseline, indicating that mitigation overhead can itself introduce additional noise at higher T-depths. While QEM provides the only currently viable approach for improving QHE performance on NISQ hardware, it alone cannot guarantee the accuracy levels required for secure encrypted computation. In the long term, integrating QEC will be essential to bridge the gap between theoretical and practical QHE security, though this remains infeasible until hardware scales beyond current NISQ limitations. Additionally, reducing auxiliary overhead through state reuse mechanisms and T-gadget compression techniques [18] represents a critical pathway toward scalable AUX-QHE implementations.

VI. Conclusion

We present the first implementation of the FHE-AUX-QHE framework on real quantum hardware, demonstrating its practical viability and identifying critical performance bottlenecks. Our results reveal that the exponential auxiliary-state increase is the primary barrier to scalability, with the 5q-3t configuration reaching 31,025 auxiliary states and 5,505 cross-terms. Hardware fidelities (F=0.12–0.38) exhibit significant noise degradation relative to the ideal simulation (F=1.0), although combined error mitigation (Opt-3+ZNE) yields consistent improvements. These results establish concrete resource thresholds for AUX-QHE deployment on NISQ devices and motivate future work on auxiliary-state optimisation and QEC integration for secure quantum cloud computing.

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